FOR EVERYONE — THE RESEARCHER, THE JOURNALIST, THE ONE WHO CHECKS

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

# Tunisia — the complete program. Everything, in one document.

Five absences. Nine legs. One metabolism. Every number recomputed in code,
every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

## 1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt \$40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for \$680M), strategic materials (phosphogypsum at Gabès costing the sea >\$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

## 2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

| LEG | TYPE | VOLUME | PRICE | \$M/YR | GRADE |
|---|---|---|---|---|---|
| Strategic metals (U + heavy REE, acid stream) | gross export | 716,500 lb $U_3O_8$ + 585 t REO | \$80-95/lb $U_3O_8$ · \$30/kg REO | \$75-86M | **STRONG** |
| Solar fuel substitution (500 MW) | import substitution | 745 GWh/yr displaced (876 GWh less 131 GWh to $H_2$) | \$142/MWh (gas \$12/MMBtu x 11 MMBtu/MWh + \$10/MWh O&M); government anchor: 1,100 GWh ≈ \$125M/yr gas saving for the same 500 MW fleet (Pg14) | \$95-116M | **PLAUSIBLE** |
| Hydrogen → ammonia (loop-internal offtake) | import substitution | 2.63 kt $H_2$/yr → 14.9 kt $NH_3$ for the GCT chain | \$500/t imported ammonia displaced (external offtake upside: \$8-9.5/kg $H_2$) | \$7-25M | *SPECULATIVE* |
| Terfas (desert truffles) | gross export | 940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported) | EUR20-40/kg farm-gate | \$14-29M | **PLAUSIBLE** |
| Food import substitution | import substitution | 1% of the \$3.0B food import bill (frame; volume bridge is a pilot deliverable) | imported protein \$15-25/kg vs local \$5-10/kg | \$30-30M | *SPECULATIVE* |
| Compute colocation services | gross export (services) | colo park, tenant-dependent | market colocation rates | \$15-15M | *SPECULATIVE* |
| ESG financing delta | financing delta (not FX) | 100-150bp on ~\$800M portfolio debt | interest saved, not foreign exchange | \$8-12M | **PLAUSIBLE** |
| **Total external-account effect (mid)** | | | | **~\$279M/yr** | **\$80M strong · \$137M plausible · \$61M speculative · \$0 verified** |

**By kind:** exports \$117M · import substitution \$152M · financing delta \$10M (interest saved, honestly labeled as not-FX). **By grade:** \$80M strong · \$137M plausible · \$61M speculative · \$0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

## 3. The energy ledger — every kWh, by leg

| LEG | GWH/YR | THE FORMULA |
|---|---|---|
| Pump-as-turbines (return flows) | 3.9 | 70% × 150k m$^3$/d × 50 m × 75% × 8,760 h |
| Heat (kiln + exotherm) | 30 | 9% of 333 GWh envelope |
| Flexibility (load shift) | 20 | dispatch value — not recovered joules |
| CO&sub2; mineralized | 30 kt | mass, not energy |
| **Total** | **33.9 recovered + 20 dispatch** | **isobaric ERD inside the 4 kWh/m$^3$ benchmark — never double-counted** |

## 4. Phase 0 — every dollar, itemized

| ITEM | COST | WHAT IT BUYS |
|---|---|---|
| Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate) | \$35,000 | the first anchor measurement |
| Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access | \$40,000 | the second anchor measurement |

| ITEM | COST | WHAT IT BUYS |
|---|---|---|
| Emissions inventory: 30 passive samplers (PM2.5/SO2/fluorides) + lab | $25,000 | baseline for the health scoreboard |
| Cohort protocol: 3,000-family registry, ethics, data platform v0 | $35,000 | the DALY ledger becomes MEASURED |
| Health pilot first tranche: 100 households + 2 schools, 90 days | $58,000 | filters, clean-air boxes, KPIs |
| Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration | $25,000 | structure before scale |
| Engineering + project management | $35,000 | the 90 days are run |
| **List price** | **$253,000** | **list price $253k; the two anchor measurements cost $75k and land first — the claimed range is the phased-execution plan, not the list** |

## 5. Every claim, graded

| CLAIM | GRADE | WHY |
|---|---|---|
| $40.5B debt (~79% GDP); 26.2% of export revenues to service | STRONG | MEASURED via World Bank API on 2026-09-09: debt $40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024) |
| $3.0B food and $4.9B fuel imports | STRONG | fuel $4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification |
| InstaDeep founded in Tunis, sold for $680M | STRONG | public deal record; the emblem for talent without infrastructure |
| Gabès: >$58M/yr of REE value lost to the sea | STRONG | Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~$58M/yr losses |
| U + heavy REE from the acid stream, ~$80M/yr | STRONG | flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0 |
| ~$285M/yr external-account effect, tiered | TIERED | $80M strong (proven flowsheet/cited anchor) · $137M plausible (gated) · $68M speculative (assumed volume/price) |
| 33.9 GWh/yr recovered + 20 GWh/yr dispatch value | DERIVED | every leg computed from cited primitives; dispatch value honestly labeled — not joules |
| $2,592 per healthy year (water rung) | **PLAUSIBLE** | arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED |
| Phase 0 = $150-250k, itemized | **PLAUSIBLE** | itemized list price $253k; the two anchor measurements cost $75k and land first |
| Commercially validated | ZERO | no buyer has signed anything — published as zero on purpose |

## 6. The hidden dependencies

| DEPENDENCY | TYPE | STATUS | NOTE |
|---|---|---|---|
| Desal brine permit | regulatory | in design | Gabes outfall / diffuser; permit required before SWRO FID |
| PG chemistry (Ra-226 partitioning) | technical | Phase 0 assay | flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream |
| Uranium handling | regulatory | gate | CNSTN licensing, NORM transport, export controls |
| REE separation capex | capital | later phase | +$244M solvent-extraction plant beyond the ~$100M circuit |
| Solar bankability | financial | gate | DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required |
| Grid connection | infrastructure | later phase | 3 GW connection + 400 kV upgrades ~$450M |
| Compute accelerators | regulatory | gate | export-control licenses; colocation until an anchor tenant is licensed |
| Truffle yield | agronomic | gate | 376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha |
| Export prices | market | exposure | U spot $80-95/lb; truffle EUR20-40/kg — no contracted offtake yet |
| Financing | financial | uncommitted | DFI channel modeled for solar/desal; nothing committed |

## 7. The structure and the pilot/capital separation

**The design is six independently bankable modules sharing infrastructure.** The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDR (2023): Tunisia's water-sector investment need ~$17.1B and energy-sector ~$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is $150-250k of measurements, and the whole core build is ~$250-650M phased over years 2-5, module by module, each gated. The CCDR numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~$31/MWh awards are the reference the DSCR gate is tested against.

## 8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **$9.34B** = 4.3 months of imports; food $3.00B + fuel $4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

**Channel 1 — the debt trap:** the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

**Channel 2 — imported inflation:** the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food ($3.00B) and fuel ($4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

**The loop’s answer:** owned export engines earn ~$117M/yr of gross exports outside the Fed cycle; the substitution legs take $152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

## The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

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## The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government's own 1,100 GWh = 5% of production, ~$125M/yr gas saving), Rc13 ($31/MWh inside the $24-35 historical bid band), Pg15 (fuel imports $4.895B, WITS-verified).

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## The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

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## The water stack — line-by-line, with each credit's failure mode

The \$0.347/m$^3$ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

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## The Darwinian ledger — why business as usual loses

Protein built the human brain (Pg27); the same lever, applied on purpose, is the plan's deepest one. The business-as-usual ledger is the fear-facts: what “keep doing what we do” actually buys.

| FACT | THE NUMBER | WHAT IT MEANS FOR THE NEXT GENERATION'S BRAINS |
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| Fluorosis belt | 24% dental fluorosis (Rw1) | high-F water ↔ lower child IQ (Pg28) — BAU taxes the cortex before it forms |
| Protein poverty | first 1,000 days (Pg29) | permanent cognitive deficit — the trees version, built child by child |
| Brain drain | tertiary emigration among world's highest (Pg30) | the country exports its cortex like its phosphate |
| The proof | InstaDeep sold for $680M | the cognitive gold was real — and sold, not kept |
| Debt | 26.2% of export revenues to service (Pg13) | the budget cannot buy the interventions the brain needs |
| The idle cohort | 28% unemployment; 54% women graduates (Rm19) | built brains, no environment — adaptation = exit |
| Inflation | 5.15% on $4.90B fuel + $3.00B food | the first cut is diet quality — the Fed reaches the children's IQ |

| SELECTION PRESSURE | WITHOUT THE PLAN | WITH THE PLAN |
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| Water scarcity | selects emigration — the skilled leave first | the mesh re-specifies the environment; adaptation becomes staying |
| Protein poverty | caps cognition permanently | ponds + canteens + quota reverse the pressure |
| The debt cycle | the surplus is eaten; the economy cannot invest | owned exports + substitution cut the invoice |
| Institutions | gates stay closed; skills emigrate through them | public gates select for competence, in public |

## The open-field plan — terfas, hectare by hectare

| RUNG | HECTARES | SEEDLINGS | PRODUCTION | VALUE | OPENS ON |
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| Gate (yrs 1-4) | 100 | 600,000 | ≥200 kg/ha | first fruit ≥€20/kg | 3-season record + contract |
| Scale (yrs 4-8) | 2,500 | 15M | 940 t/yr | €16544-29704M/yr | nursery output |
| Ceiling (yrs 8+) | 3,000 | 18M | 1,128 t/yr | inside 1,000-1,500 t/yr market cap (Rc8) | market offtake signed |

6,000 mycorrhized seedlings/ha (Rc16) — the nursery is the first asset: 5M seedlings/yr of local-ecotype capacity, ~200 nursery + ~1,000 field + ~783 seasonal jobs at scale. Local wild genetics, not the imported soil biology that failed in Saudi Arabia (Rc17).

## The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

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## The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

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## The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at $0.21 is 7× the loop's $0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 $m^3$/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

## 9. The state program — architecture, instruments, financing, adoption

### Counterparts — who signs what

| INSTITUTION | INSTRUMENT | ROLE |
|---|---|---|
| **GCT (Tunisian Chemical Group)** | acid-stream MOU + sampling agreement | the treasury's feedstock |
| **SONEDE** | municipal/industrial water offtake at >=$0.68/m3 gate | water contracts under the existing regime |
| **STEG** | 25-year PPA under the IPP regime (500 MW class) | the power leg's revenue |
| **Ministry of Industry, Mines & Energy** | program sponsor; tender registration | the executive mandate |
| **Ministry of Health** | biomonitoring + cohort registry | the scoreboard |
| **CNSTN** | NORM licensing for the uranium route | the regulatory pass |
| **Ministry of State Domains & Agriculture** | 40-year leases under Law 93-120 (86,000 ha available) | the land |
| **BCT** | reserve/rating path; FX policy coordination | the monetary endgame |

### Legal instruments — nothing new needs to be invented

| INSTRUMENT | WHAT IT PROVIDES | CITATION |
|---|---|---|
| Organic Law 2016-35 (BCT) | monetary autonomy; the program does NOT monetize | Pg4/Pg5 |
| IPP regime — Law 2015-62 & 2024/25 awards | renewables concessions; ~$31/MWh PPAs; 500 MW precedent | Rc13 |
| PPP framework — Law 2015-49 | desal/metals as PPPs where the state is the offtaker | Pg2 |
| Law 93-120 (state-domain leases) | 40-year agricultural leases; ~86,000 ha earmarked | Rc19 |
| NORM/radiological regime (CNSTN, IAEA standards) | the uranium route's license layer | Pg11/Rm12 |

### Financing stack — staged, gated, modeled in code

| STAGE | FINANCING | GATE / CONDITION |
|---|---|---|
| Phase 0 ($150-253k) | grant / founder / early mandate | no DFI needed; 90 days |
| Health pilot ($58k) | ESG/grant | the first scoreboard numbers |
| Metals plant (~$100M) | 60% debt @7% + offtake security; MIGA | DSCR >=1.2 at contract prices (nations_v5) |
| Solar (500 MW, ~$550M) | DFI stack: EIB/AfDB/WB-IFC at 4.5% + MIGA; or lean build | DSCR gate inside the bankable region (nations_v5) |
| Desal (~$165M) | 70% debt @6% + municipal offtake + sovereign backstop | blended >=$0.68/m3, >=60% pre-sold |
| Terfas + food ($7-35M) | concessional agri + founder | nursery first |
| EU Global Gateway (17 GW) | the national envelope this program plugs into | Pg7 |

### Governance

- every gate is a public number — published, dated, re-checkable
- health fund separate from the industrial SPVs; one shared data platform
- quarterly scoreboard: water chemistry, filter uptime, cohort KPIs, price index in Gafsa/Gabes
- each pilot passes or dies alone, in public — the falsification ladder is the governance
- commercially-validated ledger published at zero until the first signature moves it

### Adoption path — what the state signs, in order

1. **Month 0:** the state signs: GCT sampling MOU + Phase 0 mandate ($150-250k, 90 days, no construction)
2. **Day 90:** two anchor measurements land: the 100-point water panel and the acid-stream assays — every projection becomes a signed number
3. **Months 4-18:** pilots on public gates: acid pilot, 100 ha terfas + nursery, spirulina pond, solar bid, desal offtake, health KPIs
4. **Years 2-5:** build only what passed: metals, solar, desal, health at basin scale — module by module, each with its own financing
5. **Years 5-10+:** the full loop: municipal spine, compute colocation, hydrogen, brine valorization — the destination the original design described

## Investor essentials — the FX bridge tiered, and the pathway economics

By grade: **$80M strong** · $137M plausible · $61M speculative · $0 validated — the zero is published on purpose, and each gate is the instrument that moves it. Pathway economics, computed in code (nations_v5.py):

| PATHWAY | REVENUE | CAPEX | GROSS MARGIN | PAYBACK | DSCR |
|---|---|---|---|---|---|
| P2 metals | $75-86M/yr | ~$100M | 0-37% | 6.3 yr (mid) | 2.40 @60% debt |
| P3 solar | $27.2M/yr | ~$0M | 71% | n/a | 0.46 mkt / 0.69 DFI — clearing PPA ≥$45652/MWh |
| P4 desal | $30.1M @$0.55 → $38.3M @$0.70 | ~$0M | gate-dependent | n/a | 0.54 @$0.55 → 1.36 @$0.70 — clears |

## Engineer essentials — the numbers an EPC checks first

Energy: 33.9 GWh/yr recovered + 20 GWh dispatch — PAT 3.9, heat 30, CO&sb2; 30 kt mineralized. Water: SWRO 4 kWh/m³ lift floor 1.09 kWh/m³ per 400 m (Rw4); gate $0.68 blended, ≥60% pre-sold. Materials: Gafsa rock 322 mg/kg REE avg (Rm1); 60-85% to PG (Rm2); 85-90% of rock-U to the acid stream (Ru1), >95% recovery proven 1978-99 (Ru1); Ra-226 214-970 Bq/kg vs the 1,000 Bq/kg exemption (Rm11/Rm12); Phase 0 assays at 12 stream points.

## Health essentials — the DALY arithmetic and the pilot

$2,592 per healthy year against the $4,000/DALY standard; 10 ha spirulina feeds 178,082 children at the RCT dose; basin needs 0.7 ha. Pilot: 100 households + 2 schools, 90 days, $58,000 — gates: filter uptime ≥70%, urinary F ↓20%, results public.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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